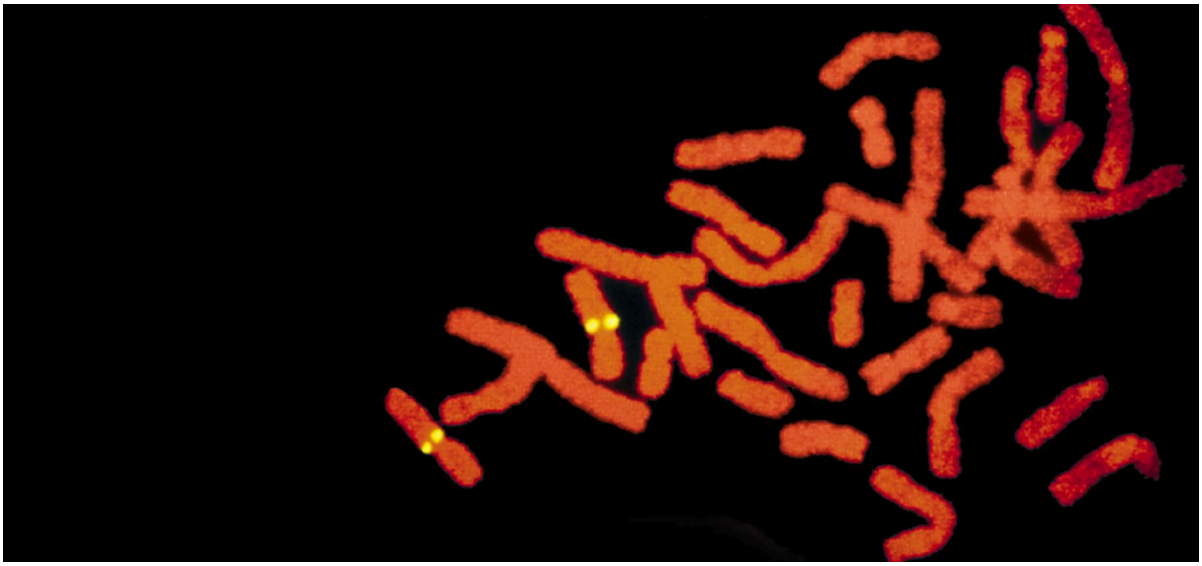
Chapter 15

Campbell Biology, AP Edition, 12e

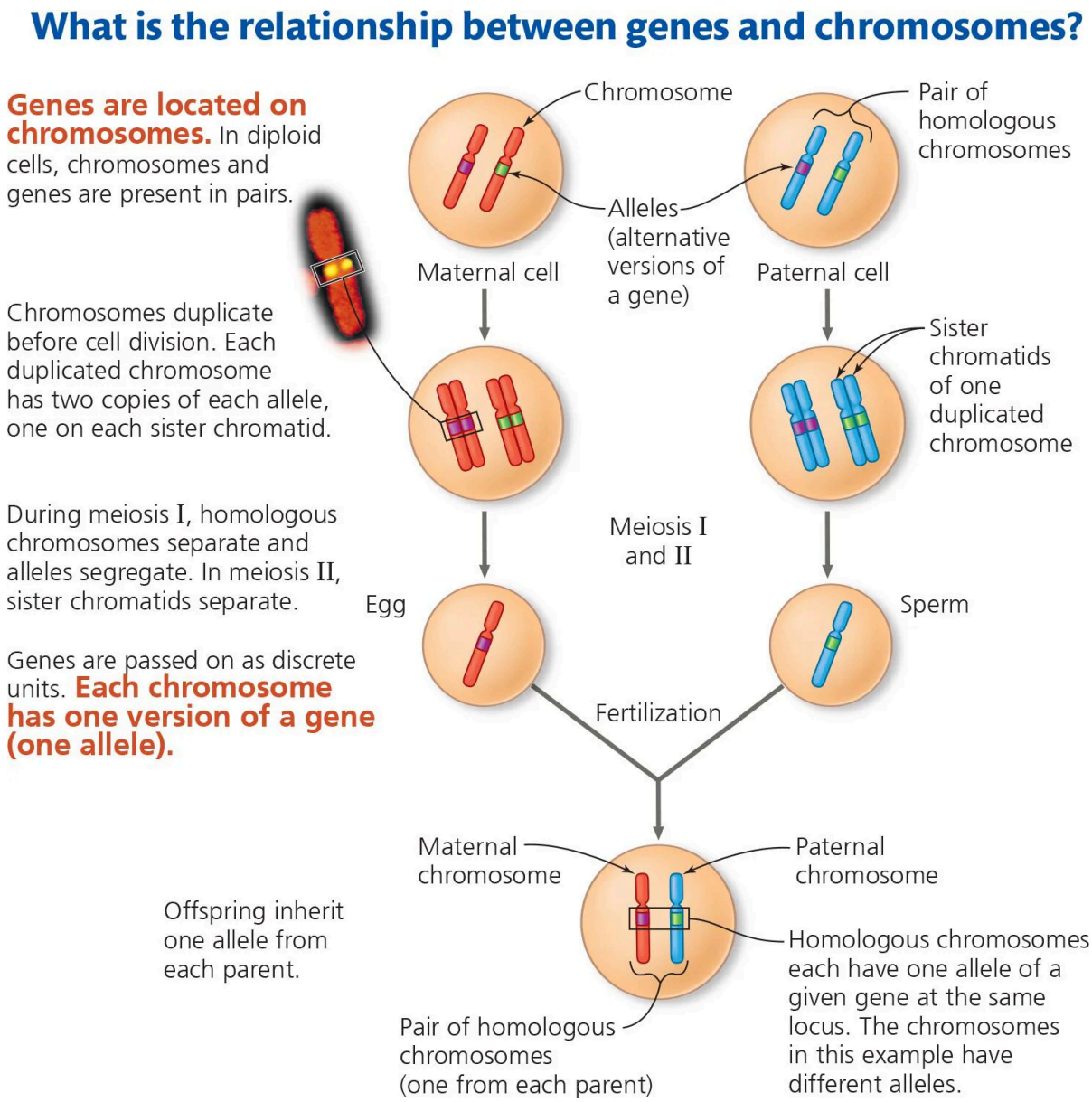


Chapter 15

The Chromosomal Basis of Inheritance

Figure 15.1

The four yellow dots mark the locations of a specific gene, tagged with a fluorescent yellow dye, on a pair of homologous chromosomes. The chromosomes have duplicated, so each chromosome has two sister chromatids, each with a copy of the gene. This provides a visual demonstration that genes—Mendel's "factors"—are segments of DNA located along chromosomes.



Genes are located on chromosomes. In diploid cells, chromosomes and genes are present in pairs. Chromosomes duplicate before cell division. Each duplicated chromosome has two copies of each allele, one on each sister chromatid. Genes are passed on as discrete units. Each chromosome has one version of a gene (one allele). A maternal cell and a paternal cell each have chromosomes. The cells go through meiosis 1 and 2, resulting in a cell with an egg and a sperm. Through fertilization, offspring inherit one allele from each parent. The offspring has a pair of homologous chromosomes. Homologous chromosomes each have one allele of a given gene at the same locus.

Key Concepts

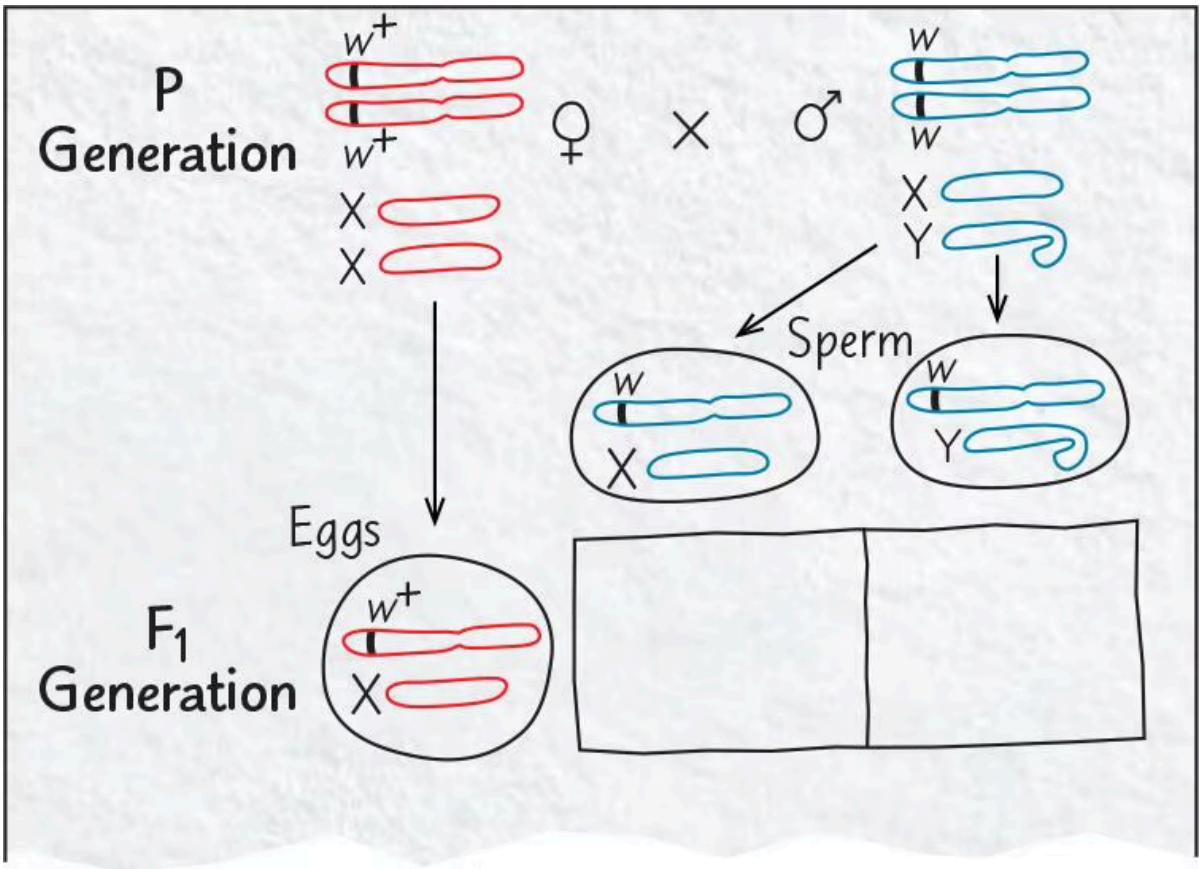
- **15.1Mendelian inheritance has its physical basis in the behavior of chromosomes****

- **15.2Sex-linked genes exhibit unique patterns of inheritance****
- **15.3Linked genes tend to be inherited together because they are located near each other on the same chromosome****
- **15.4Alterations of chromosome number or structure cause some genetic disorders****
- **15.5Some inheritance patterns are exceptions to standard Mendelian inheritance****

Segregation and independent assortment of linear chromosomes result in genetic variation **(BIG IDEA 3)**, which is structural and functional evidence that supports the relatedness of all eukaryotes **(BIG IDEA 1)**. The unique patterns of inheritance involving Barr bodies are produced by the interaction of RNA and the X chromosome **(BIG IDEA 4)**.

Study Tip

Draw chromosomes: As you work on the figure legend questions in this chapter, draw the chromosomes for the crosses described in the questions. Below is a starting sketch for the What If? question that follows Figure 15.3. (Here you are including sex chromosomes to keep track of the offspring.)



P generation has male and female chromosomes. The female chromosomes have alleles for W superscript plus on each, and the male chromosomes have alleles for w on each. X and X from the female leads to eggs with a w plus allele and an X chromosome. For the male, X and Y split into a chromosome with a w allele and an X chromosome, and a chromosome with a w allele and a Y chromosome.

Get Ready for This Chapter

In this chapter, you will learn about how genes are located on chromosomes and will explore the chromosomal basis for the transmission of genes from parents to offspring, including what happens when two genes are linked on the same chromosome. You'll also learn about sex-linked (X-linked) genes and their inheritance, sex determination, sex chromosomal abnormalities, gene imprinting, and the inheritance of genes in organelles. Before beginning this chapter, you should be able to do the following:

- Describe the process of meiosis (see Concept 13.3 and Figure 13.7).
- Describe what happens during crossing over (see Figure 13.9).
- Explain segregation and independent assortment (see Concept 14.1, Figure 14.5, and Figure 14.8).

The following questions provide a quick check of your basic knowledge in these areas.

1. A cell goes through S phase, meiosis I, and cytokinesis. As the daughter cells are beginning meiosis II, which of the following is an appropriate description of the contents of each cell?

```
Each has half the chromosomes and twice the amount of DNA as the original cell
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That's incorrect. Try again.

Incorrect. Review the following material: Homologous pairs separate during meiosis I, reducing the number of chromosome sets from two (diploid) to one (haploid). As each daughter cell begins meiosis II, it has one set of chromosomes—half the chromosomes found in the original cell. However, since each of these chromosomes was duplicated during S phase, the daughter cell has the same amount of DNA as the original cell.

```
Each has half the chromosomes and the same amount of DNA as the original cell.
```

That's correct.

Correct! If you had trouble with this question, review the following material: Homologous pairs separate during meiosis I, reducing the number of chromosome sets from two (diploid) to one (haploid). As each daughter cell begins meiosis II, it has one set of chromosomes—half the chromosomes found in the original cell. However, since each of these chromosomes was duplicated during S phase, the daughter cell has the same amount of DNA as the original cell.

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```
Each has half the chromosomes and one-fourth the amount of DNA as the original cell
```

That's incorrect. Try again.

Incorrect. Review the following material: Homologous pairs separate during meiosis I, reducing the number of chromosome sets from two (diploid) to one (haploid). As each daughter cell begins meiosis II, it has one set of chromosomes—half the chromosomes found in the original cell. However, since each of these

chromosomes was duplicated during S phase, the daughter cell has the same amount of DNA as the original cell.

2. During a single crossover event, how many strands of DNA must break? (Recall that DNA is double-stranded.)

```
four strands
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That's correct.

Correct! If you had trouble with this question, review the following material: After interphase, the chromosomes have been duplicated and the two members of the homologous pair associate along their length. The double-stranded DNA molecules of two nonsister chromatids—one maternal and one paternal—are broken at precisely corresponding points. During synapsis, each of the two double-stranded DNA breaks (leaving foursis closed up so that each broken (double-stranded) end is joined to the corresponding broken (double-stranded) end of the nonsister chromatid, producing crossovers.

```
eight strands
```

That's incorrect. Try again.

Incorrect. Review the following material: After interphase, the chromosomes have been duplicated and the two members of the homologous pair associate along their length. The double-stranded DNA molecules of two nonsister chromatids—one maternal and one paternal—are broken at precisely corresponding points. During synapsis, each of the two double-stranded DNA breaks (leaving four broken ends) is closed up so that each broken (double-stranded) end is joined to the corresponding broken (double-stranded) end of the nonsister chromatid, producing crossovers.

```
one strand
```

That's incorrect. Try again.

Incorrect. Review the following material: After interphase, the chromosomes have been duplicated and the two members of the homologous pair associate along their length. The double-stranded DNA molecules of two nonsister chromatids—one maternal and one paternal—are broken at precisely corresponding points. During synapsis, each of the two double-stranded DNA breaks (leaving four broken ends) is closed up so that each broken (double-stranded) end is joined to the corresponding broken (double-stranded) end of the nonsister chromatid, producing crossovers.

```
        two strands
```

That's incorrect. Try again.

Incorrect. Review the following material: After interphase, the chromosomes have been duplicated and the two members of the homologous pair associate along their length. The double-stranded DNA molecules of two nonsister chromatids—one maternal and one paternal—are broken at precisely corresponding points. During synapsis, each of the two double-stranded DNA breaks (leaving four broken ends) is closed up so that each broken (double-stranded) end is joined to the corresponding broken (double-stranded) end of the nonsister chromatid, producing crossovers.

3. A sexually reproducing animal has two unlinked genes, one for head shape (H) and one for tail length (T). Its genotype is HhTt. Which of the following genotypes for these two genes is possible in a gamete from this organism?

** Tt**

That's incorrect. Try again.

Incorrect. Review the following material: Mendel’s law of segregation states that the two alleles for each heritable character (in this case, head shape or tail length) segregate during gamete formation and end up in different gametes. The law of independent assortment states that each pair of alleles from unlinked genes will assort independently of any other allele pair during gamete formation. Therefore, each gamete will receive one allele for head shape and one allele for tail length, resulting in the following possible genotypes of gametes produced by this organism: HT,hT,Ht, and ht.

** HT**

That's correct.

Correct! If you had trouble with this question, review the following material: Mendel’s law of segregation states that the two alleles for each heritable character (in this case, head shape or tail length) segregate during gamete formation and end up in different gametes. The law of independent assortment states that each pair of alleles from unlinked genes will assort independently of any other allele pair during gamete formation. Therefore, each gamete will receive one allele for head shape and one allele for tail length, resulting in the following possible genotypes of gametes produced by this organism: HT,hT,Ht, and ht.

HhTt

That's incorrect. Try again.

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T

That's incorrect. Try again.

Incorrect. Review the following material: Mendel’s law of segregation states that the two alleles for each heritable character (in this case, head shape or tail length) segregate during gamete formation and end up in different gametes. The law of independent assortment states that each pair of alleles from unlinked genes will assort independently of any other allele pair during gamete formation. Therefore, each gamete will receive one allele for head shape and one allele for tail length, resulting in the following possible genotypes of gametes produced by this organism: HT,hT,Ht, and ht.

Learning Objectives

After studying this chapter, you should be able to do the following:

- 15.1 Discuss Mendel's laws of inheritance in relation to the behavior of chromosomes during meiosis and fertilization, and explain Morgan's experiments with red- and white-eyed fruit flies.
- 15.2 Describe the inheritance patterns displayed by sex-linked genes and the effects of X inactivation in female mammals.
- 15.3 Explain how linkage of genes and crossing over affect inheritance.
- 15.4 List different types of alterations to chromosome number and structure, and explain how they can cause genetic disorders.
- 15.5 Identify the basis for non-Mendelian inheritance patterns shown by (1) organelle genes and (2) imprinted genes.